Power for Two-Proportion Test

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Introduction

Comparing proportions between two independent groups is one of the most common designs in clinical research, public-health epidemiology, and quality-control. Whenever the primary outcome is binary — event vs no event, response vs non-response, success vs failure — the trial's sample-size calculation reduces to a two-proportion power analysis. Several closely-related approaches are widely used: Cohen's arcsine-transformed effect size h that maps cleanly to a Normal-approximation formula, direct sample-size formulas in the original-proportion scale (e.g., Kelsey's formula with continuity correction), and exact Fisher-based methods for rare events. The choice depends on the expected event rates, the desired allocation ratio, and the regulatory or methodological context.

Prerequisites

A working understanding of the two-proportion test, the arcsine variance-stabilising transformation, and Cohen's h as the standardised effect-size measure for proportion-difference designs.

Theory

Cohen's h is the arcsine-transformed proportion difference

$$h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2}.$$

For balanced two-arm designs, the sample size per group at two-sided α and power $1 - \beta$ is approximately

$$n \approx 2 \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{h} \right)^2 \text{ per group.}$$

For unequal allocation with ratio $k = n_2/n_1$, the total sample size grows by a factor $(1 + 1/k)(1 + k)/4$ relative to the balanced minimum. Cohen's benchmarks for h are 0.20 (small), 0.50 (medium), and 0.80 (large).

Assumptions

The two groups are independent, the standard Normal-approximation conditions $np \geq 10$ and $n(1 - p) \geq 10$ are satisfied for each group, and the proportions under the alternative hypothesis are pre-specified.

R Implementation

```r
library(pwr); library(Hmisc)

pwr.2p.test(h = ES.h(0.20, 0.10), sig.level = 0.05, power = 0.80)

bsamsize(p1 = 0.20, p2 = 0.10, alpha = 0.05, power = 0.80)
```

```r
pwr.2p2n.test(h = ES.h(0.20, 0.10), n1 = 100, sig.level = 0.05)
n1_fix <- 100
n2_vals <- seq(40, 300, 10)
pw <- sapply(n2_vals, function(n2)
  pwr.2p2n.test(h = ES.h(0.20, 0.10), n1 = n1_fix, n2 = n2)$power)
n2_vals[which(pw >= 0.80)[1]]
```

Output & Results

`pwr.2p.test()` returns the per-group sample size at specified h, α, and power; `Hmisc::bsamsize()` provides the same calculation directly in the original-proportion scale. For $p_1 = 0.20$ vs $p_2 = 0.10$ at 80 % power, $n \approx 199$ per group (398 total). The unequal-allocation analysis shows that fixing $n_1 = 100$ requires roughly $n_2 = 280$ to maintain 80 % power, illustrating the inefficiency of imbalanced designs.

Interpretation

A reporting sentence: "To detect a difference in event rates from 10 % (historical control) to 20 % (active treatment) at $\alpha = 0.05$ two-sided with 80 % power, 199 patients per arm are required (398 total) under a 1:1 allocation. The protocol enrols 440 patients to allow for 10 % attrition. Sensitivity analyses across $p_2 \in [0.15, 0.25]$ yield required per-arm n from 392 to 105, bracketing the assumed effect; the conservative end of this range was used to justify the operational sample size." Always report the assumed proportions and the allocation ratio.

Practical Tips

- Equal allocation (1:1) is most efficient for detecting a difference between two proportions; unequal allocation (2:1, 3:1) costs 10–30 % more total n for the same power and is justified only when the cost or feasibility of the two arms differs substantially.
- For rare events ($p < 0.01$ in either arm), the Normal-approximation formulas are unreliable; use exact Fisher's-test-based power calculations or simulation, available in `Exact::power.exact.test()` or via Monte Carlo enumeration.
- Continuity correction (Yates) inflates the required sample size slightly and is the historical default for 2×2 chi-squared tests; modern practice often omits it for likelihood-ratio or score tests, so verify what your software is computing.
- For cluster-randomised designs, inflate the calculated n by the design effect $1 + (\bar{m} - 1)\text{ICC}$, where $\bar{m}$ is the average cluster size and ICC is the intraclass correlation; ignoring clustering grossly under-powers cluster trials.
- Cohen's h benchmarks (0.20 / 0.50 / 0.80) are useful for translating effect-size language into sample-size expectations, but tying the calculation to the actual expected proportions via `ES.h(p1, p2)` is more defensible than invoking the benchmarks directly.
- For non-inferiority or equivalence designs, the calculation differs from a superiority test; use `pwr.2p.test()` with the appropriate margin or specialised tools like `bsamsize()` with non-inferiority arguments.

R Packages Used

`pwr::pwr.2p.test()` and `pwr::pwr.2p2n.test()` for canonical Cohen's h Normal-approximation calculations; `Hmisc::bsamsize()` for direct-proportion-scale calculations with optional continuity correction; `Exact::power.exact.test()` for exact-binomial power across small-sample two-proportion designs; `clusterPower` for cluster-randomised two-proportion power; `Mediana` for trial-design simulation including two-proportion power across complex designs.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation